

Unit 11 (Part 1)

The Eigenvalue Decomposition

Unit Goals

- Be able to define what is meant by an eigenvalue of a matrix.
- Be able to solve for the eigenvalues of 2-by-2 and 3-by-3 matrices.
- Be able to solve for the eigenvalues of certain n -by- n matrices with characteristic forms.
- Be able define what is meant by an eigenvector and an eigenspace.
- Be able to find a linearly independent set of eigenvectors associated with a given eigenvalue of a matrix.
- Be able to diagonalize a matrix, or determine that it can not be diagonalized, given the eigenvalues.
- Be able to use the eigenvalue decomposition to solve real-world problems.

Unit 11.1

**What are eigenvalues and
eigenvectors?**

So far ... $A\vec{x} = \vec{b}$.

- Consider problems involving
 - balancing production with demand,
 - production with available resources,
 - economic output, and
 - chemical equations
- These problems gave rise to systems of equations that could be expressed as:

$$A\vec{x} = \vec{b}$$

So far ... $A\vec{x} = \vec{b}$.

- We can solve these exactly with **elimination** or approximately with **projections**
- These ideas were captured in the **LU** and **QR** decompositions.
- Both decompositions allow us to perform one “expensive” calculation on A , then solve for any number of right-hand sides with **little effort**:

$$PA = LU \rightarrow \begin{aligned} L\vec{c} &= P\vec{b} \\ U\vec{x} &= \vec{c} \end{aligned}$$

$$A = QR \rightarrow R\vec{x} = Q^T\vec{b}$$

Markov Chains and Dynamical Systems

- Markov processes and dynamical systems fundamentally give rise to systems of equations of the form:

$$A\vec{x}_i = \vec{x}_{i+1}$$

- We can find stationary values by setting $\vec{x}_i = \vec{x}_{i+1}$, which gives rise the homogenous system $(I - A)\vec{x} = \vec{0}$, but what about the more general case?

Markov Chains and Dynamical Systems

- Can we find one “expensive” calculation that would make the evaluation of a sequence of $\vec{x}_i$ much cheaper?
- We want to avoid matrix multiplication.
- Begin by considering a special case:

$$\begin{aligned} A\vec{x} &= k\vec{x} \rightarrow \\ \vec{x}_{i+1} &= A\vec{x}_i = k\vec{x}_i \rightarrow \\ \vec{x}_{i+n} &= A^n\vec{x}_i = k^n\vec{x}_i \end{aligned}$$

What are eigenvalues and eigenvectors?

Definition:

- If a scalar λ and a non-zero vector $\vec{x}$ satisfy:

$$A\vec{x} = \lambda\vec{x}$$

- Then we say that λ is an **eigenvalue** of A , and $\vec{x}$ is an **associated eigenvector**.

What are eigenvalues and eigenvectors?

Notes:

- When we multiply an eigenvector by A (on the left), the result is just a **scalar multiple** of the eigenvector.
- The scaling factor is the **eigenvalue**.
- **Eigenvectors** are always associated with some **eigenvalue**.

How do eigenvalues and eigenvectors help?

- If we can express an arbitrary vector as a linear combination of eigenvectors, matrix multiplication becomes easy!

$$\vec{u} = a_1\vec{x}_1 + a_2\vec{x}_2 + \cdots, \quad A\vec{x}_1 = \lambda_1\vec{x}_1, A\vec{x}_2 = \lambda_2\vec{x}_2, \cdots$$

$$\begin{aligned} A\vec{u} &= A(a_1\vec{x}_1 + a_2\vec{x}_2 + \cdots) \\ &= a_1A\vec{x}_1 + a_2A\vec{x}_2 + \cdots \\ &= a_1\lambda_1\vec{x}_1 + a_2\lambda_2\vec{x}_2 + \cdots \end{aligned}$$

$$A^k\vec{u} = a_1(\lambda_1)^k\vec{x}_1 + a_2(\lambda_2)^k\vec{x}_2 + \cdots$$

- **Multiplication of any power of A is replaced by sum of scalar multiplications.**

How do eigenvalues and eigenvectors help?

$$A^k \vec{u} = a_1(\lambda_1)^k \vec{x}_1 + a_2(\lambda_2)^k \vec{x}_2 + \dots$$

- When we multiply by large powers of A , only the largest (in absolute value) eigenvalue matters:

$$|\lambda_1| > |\lambda_2| > |\lambda_3| \dots \rightarrow |(\lambda_1)^k| \gg |(\lambda_2)^k| \gg |(\lambda_3)^k| \dots \text{ for } k \gg 1$$

$$A^k \vec{u} \approx a_1(\lambda_1)^k \vec{x}_1, \quad k \gg 1$$

$$\lim_{k \rightarrow \infty} A^k \vec{u} = \lim_{k \rightarrow \infty} a_1(\lambda_1)^k \vec{x}_1$$

- **Note:** This implies that $\lim_{k \rightarrow \infty} A^k \vec{u}$ can only be well defined if **all** eigenvalues (abs value) are **< 1**.

Questions.

- Can we always write a vector as a linear combination of eigenvectors?
- Do all matrices have eigenvalues and eigenvectors?
- If matrix has eigenvalues/eigenvectors, how many are there?
- How do we find them?

Finding eigenvalues.

- Is there a scalar λ and a non-zero vector $\vec{x}$ that satisfy $A\vec{x} = \lambda\vec{x}$?

$$A\vec{x} = \lambda\vec{x}$$

$$\lambda\vec{x} - A\vec{x} = \vec{0}$$

$$(\lambda I - A)\vec{x} = \vec{0}$$

- A must be square $\rightarrow$ **only square matrices have eigenvalues.**
- λ is an eigenvalue of A if and only if $(\lambda I - A)$ is singular:

$$\det(\lambda I - A) = 0$$

Finding eigenvalues.

- $\det(\lambda I - A)$ is called the **characteristic polynomial** of A ; its roots are the eigenvalues:
 - For an $n \times n$ matrix, it is a polynomial of degree n .
 - A polynomial of degree n can have at most n roots.
 - **An $n \times n$ matrix can have at most n eigenvalues.**

Finding eigenvalues.

- **Example:** What are the eigenvalues of $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$?

$$\det(\lambda I - A) = \begin{vmatrix} \lambda - 1 & -2 \\ -2 & \lambda - 1 \end{vmatrix} = (\lambda - 1)^2 - 4$$

$$= \lambda^2 - 2\lambda + 1 - 4 = \lambda^2 - 2\lambda - 3 = (\lambda - 3)(\lambda + 1)$$

$$\det(\lambda I - A) = 0 \rightarrow (\lambda - 3)(\lambda + 1) = 0 \rightarrow \lambda = 3, -1$$

Finding eigenvalues.

- **Example:** What are the eigenvalues of $A = \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix}$?

$$\det(\lambda I - A) = \begin{vmatrix} \lambda & -1 \\ +1 & \lambda - 2 \end{vmatrix} = \lambda(\lambda - 2) - (-1)$$

$$= \lambda^2 - 2\lambda + 1 = (\lambda - 1)^2$$

$$\det(\lambda I - A) = 0 \rightarrow (\lambda - 1)^2 = 0 \rightarrow \lambda = 1$$

Finding eigenvalues.

Quadratic formula:

$$ax^2 + bx + c = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- **Example:** What are the eigenvalues of $A = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}$?

$$\det(\lambda I - A) = \begin{vmatrix} \lambda - 1 & -2 \\ +2 & \lambda - 1 \end{vmatrix} = (\lambda - 1)^2 - (-4)$$

$$= \lambda^2 - 2\lambda + 1 + 4 = \lambda^2 - 2\lambda + 5$$

$$\det(\lambda I - A) = 0 \rightarrow \lambda^2 - 2\lambda + 5 = 0 \rightarrow \lambda = \frac{2 \pm \sqrt{4 - 20}}{2}$$

$$\lambda = \frac{2 \pm \sqrt{-16}}{2} = 1 \pm 2i$$

The characteristic polynomial

For an $n \times n$ matrix, $\det(\lambda I - A)$ is a polynomial of degree n .

$$\det(\lambda I - A) = \begin{vmatrix} \lambda - a_{11} & -a_{12} & \cdots & -a_{1n} \\ -a_{21} & \lambda - a_{22} & & \\ \vdots & & \ddots & \\ -a_{n1} & & & \lambda - a_{nn} \end{vmatrix}$$

- **Leading term is the product of diagonal entries:**

$$\det(\lambda I - A) = (\lambda - a_{11})(\lambda - a_{22}) \cdots (\lambda - a_{nn}) + \cdots$$

- **Highest order term is λ^n :**

$$\det(\lambda I - A) = \lambda^n + c_1 \lambda^{n-1} + \cdots + c_{n-1} \lambda + c_n$$

The characteristic polynomial

- For an $n \times n$ matrix, $\det(\lambda I - A)$ is a polynomial of degree n .
 - A polynomial of degree n can have **at most n roots**
 - Some roots may be repeated
 - Some roots may be imaginary or **complex numbers**
 - Every polynomial has at least one root when we allow for complex numbers

The characteristic polynomial

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \rightarrow \det(\lambda I - A) = (\lambda - 3)(\lambda + 1) = 0 \rightarrow \lambda = 3, -1$$

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix} \rightarrow \det(\lambda I - A) = (\lambda - 1)^2 = 0 \rightarrow \lambda = 1$$

$$A = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix} \rightarrow \det(\lambda I - A) = \lambda^2 - 2\lambda + 5 = 0 \rightarrow \lambda = 1 \pm 2i$$

- **An $n \times n$ matrix can have at most n distinct eigenvalues.**
- **Every $n \times n$ matrix has at least one eigenvalue, but they may be complex.**

Finding eigenvectors.

- If λ is an eigenvalue, then $\det(\lambda I - A) = 0$ and $(\lambda I - A)$ is **singular**;
- There must be a **non-zero vector** $\vec{x}$ that satisfies $A\vec{x} = \lambda\vec{x}$.
$$A\vec{x} = \lambda\vec{x} \rightarrow (\lambda I - A)\vec{x} = \vec{0}$$
- This is a **homogenous system** whereby solutions are described by the **null space**.

Finding eigenvectors.

- Every vector in the null space of $(\lambda I - A)$ is an **eigenvector** of A with corresponding **eigenvalue** $\lambda \rightarrow$ an **eigenspace**.
- We characterize a vector space with a basis; a **basis** is given by the special solutions to:

$$(\lambda I - A)\vec{x} = \vec{0}$$

- **Important:** Each distinct λ has a distinct **eigenspace**; we typically aim to find a **basis** for each **eigenspace**.

Finding eigenvectors.

- **Example:** Find a basis for the eigenspaces of $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$.

$$\det(\lambda I - A) = (\lambda - 3)(\lambda + 1) = 0 \rightarrow \lambda = 3, -1$$

$$\lambda = 3 \rightarrow (\lambda I - A) = \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & -2 \\ 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} = R \rightarrow \vec{x}_n = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\lambda = -1 \rightarrow (\lambda I - A) = \begin{bmatrix} -2 & -2 \\ -2 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 2 \\ 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} = R \rightarrow \vec{x}_n = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

- $\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$ is a basis for the eigenspace associated with $\lambda = 3$.
- $\left\{ \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right\}$ is a basis for the eigenspace associated with $\lambda = -1$.
- These are **linearly independent vectors** in $\mathbb{R}^2$; together they span the whole space.

Finding eigenvectors.

- **Example:** Find a basis for the eigenspaces of $A = \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix}$
 $\det(\lambda I - A) = (\lambda - 1)^2 = 0 \rightarrow \lambda = 1$

$$\lambda = 1 \rightarrow (\lambda I - A) = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} = R \rightarrow \vec{x}_n = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

- $\left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$ is a basis for the **eigenspace** associated with $\lambda = 1$.
- The eigenspace associated with the only **distinct eigenvector** has dimension 1 (it is a line); it does not span the whole space of $\mathbb{R}^2$.

Finding eigenvectors.

- **Example:** Find a basis for the eigenspaces of $A = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}$.

$$\det(\lambda I - A) = \lambda^2 - 2\lambda + 5 = 0 \rightarrow \lambda = 1 \pm 2i$$

$$\lambda = 1 + 2i \rightarrow (\lambda I - A) = \begin{bmatrix} 2i & -2 \\ 2 & 2i \end{bmatrix} \rightarrow \begin{bmatrix} 2i & -2 \\ 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & i \\ 0 & 0 \end{bmatrix} = R \rightarrow \vec{x}_n = \begin{bmatrix} -i \\ 1 \end{bmatrix}$$

$$\lambda = 1 - 2i \rightarrow (\lambda I - A) = \begin{bmatrix} -2i & -2 \\ 2 & -2i \end{bmatrix} \rightarrow \begin{bmatrix} -2i & -2 \\ 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -i \\ 0 & 0 \end{bmatrix} = R \rightarrow \vec{x}_n = \begin{bmatrix} i \\ 1 \end{bmatrix}$$

- $\left\{ \begin{bmatrix} -i \\ 1 \end{bmatrix} \right\}$ is a basis for the eigenspace associated with $\lambda = 1 + 2i$.
- $\left\{ \begin{bmatrix} i \\ 1 \end{bmatrix} \right\}$ is a basis for the eigenspace associated with $\lambda = 1 - 2i$.
- **When we have complex eigenvalues, we get complex eigenvectors!**

In summary ...

- Every square matrix has at least one eigenvalue and linearly independent eigenvector.
- An $n \times n$ matrix has at most n eigenvalues and at most n linearly independent eigenvectors.
- The solutions to $\det(\lambda I - A) = 0$ give us the eigenvalues of A .
- For each eigenvalue λ , the null space of $(\lambda I - A)$ is the eigenspace associated with λ .
- We find a basis by finding the special solutions of $(\lambda I - A)\vec{x} = \vec{0}$.

Unit 11.1.1

Worked examples: Finding the eigenvalues and eigenvectors of a matrix.

Find the eigenvalues and eigenvectors.

- Find the **eigenvalues** and a **basis** for each eigenspace for the matrix:

$$A = \begin{bmatrix} 3 & 0 & 3 \\ 0 & 6 & 0 \\ 3 & 0 & 3 \end{bmatrix}$$

Find the eigenvalues and eigenvectors.

- First find the roots of $|\lambda I - A|$ (the eigenvalues):

$$A = \begin{bmatrix} 3 & 0 & 3 \\ 0 & 6 & 0 \\ 3 & 0 & 3 \end{bmatrix} \rightarrow |\lambda I - A| = \begin{vmatrix} \lambda - 3 & 0 & -3 \\ 0 & \lambda - 6 & 0 \\ -3 & 0 & \lambda - 3 \end{vmatrix}$$

$$= (\lambda - 3) \begin{vmatrix} \lambda - 6 & 0 \\ 0 & \lambda - 3 \end{vmatrix} - 0 + (-3) \begin{vmatrix} 0 & \lambda - 6 \\ -3 & 0 \end{vmatrix}$$

$$= (\lambda - 3)((\lambda - 3)(\lambda - 6) - 0) - 3(0 - (-3)(\lambda - 6)) = (\lambda - 3)^2(\lambda - 6) - 9(\lambda - 6)$$

$$(\lambda - 6)((\lambda - 3)^2 - 9) = (\lambda - 6)(\lambda^2 - 6\lambda + 9 - 9) = (\lambda - 6)(\lambda^2 - 6\lambda) = \lambda(\lambda - 6)^2$$

$$|\lambda I - A| = 0 \rightarrow \lambda(\lambda - 6)^2 = 0 \rightarrow \lambda = 0, 6$$

Find the eigenvalues and eigenvectors.

- Then find a basis for the null space of $(\lambda I - A)$ (the eigenspace) for each λ :

$$\lambda = 0, A = \begin{bmatrix} 3 & 0 & 3 \\ 0 & 6 & 0 \\ 3 & 0 & 3 \end{bmatrix} \rightarrow (\lambda I - A) = \begin{bmatrix} -3 & 0 & -3 \\ 0 & -6 & 0 \\ -3 & 0 & -3 \end{bmatrix}$$

$$\begin{bmatrix} -3 & 0 & -3 \\ 0 & -6 & 0 \\ -3 & 0 & -3 \end{bmatrix} \rightarrow \begin{bmatrix} -3 & 0 & -3 \\ 0 & -6 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \vec{x}_{n1} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$\lambda = 0 \rightarrow \left\{ \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Find the eigenvalues and eigenvectors.

- Then find a basis for the null space of $(\lambda I - A)$ (the eigenspace) for each λ :

$$\lambda = 6, A = \begin{bmatrix} 3 & 0 & 3 \\ 0 & 6 & 0 \\ 3 & 0 & 3 \end{bmatrix} \rightarrow (\lambda I - A) = \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}$$

Find the eigenvalues and eigenvectors.

- Find the **eigenvalues** and a **basis** for each eigenspace for the matrix:

$$A = \begin{bmatrix} 3 & -1 & 1 \\ 7 & -5 & 1 \\ 6 & -6 & 2 \end{bmatrix}$$

Find the eigenvalues and eigenvectors.

- First find the roots of $|\lambda I - A|$ (the eigenvalues):

$$\begin{aligned} A &= \begin{bmatrix} 3 & -1 & 1 \\ 7 & -5 & 1 \\ 6 & -6 & 2 \end{bmatrix} \rightarrow |\lambda I - A| = \begin{vmatrix} \lambda - 3 & 1 & -1 \\ -7 & \lambda + 5 & -1 \\ -6 & 6 & \lambda - 2 \end{vmatrix} \\ &= (\lambda - 3) \begin{vmatrix} \lambda + 5 & -1 \\ 6 & \lambda - 2 \end{vmatrix} - 1 \begin{vmatrix} -7 & -1 \\ -6 & \lambda - 2 \end{vmatrix} + (-1) \begin{vmatrix} -7 & \lambda + 5 \\ -6 & 6 \end{vmatrix} \\ &= (\lambda - 3)((\lambda + 5)(\lambda - 2) - (-6)) - 1((-7)(\lambda - 2) - 6) - 1(-42 - (-6)(\lambda + 5)) \\ &= (\lambda - 3)(\lambda^2 + 3\lambda - 10 + 6) - 1(-7\lambda + 14 - 6) - 1(-42 + 6\lambda + 30) \\ &= (\lambda - 3)(\lambda^2 + 3\lambda - 4) - 1(-7\lambda + 8) - 1(6\lambda - 12) \\ &= \lambda^3 + 3\lambda^2 - 4\lambda - 3\lambda^2 - 9\lambda + 12 + 7\lambda - 8 - 6\lambda + 12 \\ &= \lambda^3 - 12\lambda + 16 \end{aligned}$$

Find the eigenvalues and eigenvectors.

$$A = \begin{bmatrix} 3 & -1 & 1 \\ 7 & -5 & 1 \\ 6 & -6 & 2 \end{bmatrix} \rightarrow |\lambda I - A| = \lambda^3 - 12\lambda + 16$$

- **If there are integer solutions, they must be factors of the last term: $\pm 2, \pm 4, \pm 8, \pm 16$**

$$2^3 - 12(2) + 16 = 8 - 24 + 16 = 0 \rightarrow \lambda = 2 \rightarrow \lambda - 2 = 0$$

- **Keep testing or factor out the term you have found:**

$$\lambda^3 - 12\lambda + 16 = (\lambda - 2)(\lambda^2 + a\lambda + b) \rightarrow -2b = 16 \rightarrow b = -8$$

$$\lambda^3 - 12\lambda + 16 = (\lambda - 2)(\lambda^2 + a\lambda - 8) \rightarrow -8\lambda - 2a\lambda = -12\lambda \rightarrow a = 2$$

$$\lambda^3 - 12\lambda + 16 = (\lambda - 2)(\lambda^2 + 2\lambda - 8) = (\lambda - 2)(\lambda + 4)(\lambda - 2) = (\lambda - 2)^2(\lambda + 4)$$

$$|\lambda I - A| = 0 \rightarrow (\lambda - 2)^2(\lambda + 4) = 0 \rightarrow \lambda = 2, -4$$

Find the eigenvalues and eigenvectors.

- Then find a basis for the null space of $(\lambda I - A)$ (the eigenspace) for each λ :

$$\lambda = -4, A = \begin{bmatrix} 3 & -1 & 1 \\ 7 & -5 & 1 \\ 6 & -6 & 2 \end{bmatrix} \rightarrow (\lambda I - A) = \begin{bmatrix} -7 & 1 & -1 \\ -7 & 1 & -1 \\ -6 & 6 & -6 \end{bmatrix}$$

$$\begin{bmatrix} -7 & 1 & -1 \\ -7 & 1 & -1 \\ -6 & 6 & -6 \end{bmatrix} \rightarrow \begin{bmatrix} -7 & 1 & -1 \\ 0 & 0 & 0 \\ 1 & -1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 1 \\ -7 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 1 \\ 0 & -6 & 6 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow$$

$$\rightarrow \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \vec{x}_{n1} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \lambda = -4 \rightarrow \left\{ \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$$

Find the eigenvalues and eigenvectors.

- Then find a basis for the null space of $(\lambda I - A)$ (the eigenspace) for each λ :

$$\lambda = 2, A = \begin{bmatrix} 3 & -1 & 1 \\ 7 & -5 & 1 \\ 6 & -6 & 2 \end{bmatrix} \rightarrow (\lambda I - A) = \begin{bmatrix} -1 & 1 & -1 \\ -7 & 7 & -1 \\ -6 & 6 & 0 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 & -1 \\ -7 & 7 & -1 \\ -6 & 6 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 1 & -1 \\ 0 & 0 & 6 \\ 0 & 0 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 1 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \vec{x}_{n1} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \lambda = 2 \rightarrow \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\}$$

Find the eigenvalues and eigenvectors.

- Find the eigenvalues and a basis for each eigenspace for the matrix:

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 2 & 2 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

Find the eigenvalues and eigenvectors.

- First find the roots of $|\lambda I - A|$ (the eigenvalues):

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 2 & 2 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 4 \end{bmatrix} \rightarrow |\lambda I - A| = \begin{vmatrix} \lambda - 1 & 1 & 1 & 1 \\ 0 & \lambda - 2 & 2 & 2 \\ 0 & 0 & \lambda - 3 & 3 \\ 0 & 0 & 0 & \lambda - 4 \end{vmatrix}$$
$$= (\lambda - 1)(\lambda - 2)(\lambda - 3)(\lambda - 4)$$

$$|\lambda I - A| = 0 \rightarrow (\lambda - 1)(\lambda - 2)(\lambda - 3)(\lambda - 4) = 0 \rightarrow \lambda = 1, 2, 3, 4$$

- **Note: The eigenvalue of an upper (or lower) triangular matrix are simply the diagonal entries!**

Find the eigenvalues and eigenvectors.

- Find a basis for the null space of $(\lambda I - A)$ (the eigenspace) for each λ :

$$\lambda = 1, A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 2 & 2 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 4 \end{bmatrix} \rightarrow (\lambda I - A) = \begin{bmatrix} 0 & -1 & -1 & -1 \\ 0 & -1 & -2 & -2 \\ 0 & 0 & -2 & -3 \\ 0 & 0 & 0 & -3 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 & -1 & -1 \\ 0 & -1 & -2 & -2 \\ 0 & 0 & -2 & -3 \\ 0 & 0 & 0 & -3 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & -1 & -1 & 0 \\ 0 & -1 & -2 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & -1 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\rightarrow \vec{x}_{n1} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \lambda = 1 \rightarrow \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \right\}$$

Find the eigenvalues and eigenvectors.

- Then find a basis for the null space of $(\lambda I - A)$ (the eigenspace) for each λ :

$$\lambda = 2, A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 2 & 2 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 4 \end{bmatrix} \rightarrow (\lambda I - A) = \begin{bmatrix} 1 & -1 & -1 & -1 \\ 0 & 0 & -2 & -2 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & -1 & -1 \\ 0 & 0 & -2 & -2 \\ 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & -1 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\rightarrow \vec{x}_{n1} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \lambda = 2 \rightarrow \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \right\}$$

Find the eigenvalues and eigenvectors.

- Then find a basis for the null space of $(\lambda I - A)$ (the eigenspace) for each λ :

$$\lambda = 3, A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 2 & 2 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 4 \end{bmatrix} \rightarrow (\lambda I - A) = \begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix}$$

Find the eigenvalues and eigenvectors.

- Then find a basis for the null space of $(\lambda I - A)$ (the eigenspace) for each λ :

$$\lambda = 4, A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 2 & 2 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 4 \end{bmatrix} \rightarrow (\lambda I - A) = \begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix}$$